

MUHAMMAD AKMAL

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SUMMARY

Computer Science graduate with a strong foundation in Python, FastAPI, **Machine Learning**, and **Artificial Intelligence**. Experienced in building intelligent solutions, developing modern web applications, and solving real-world problems. Passionate about building scalable **AI applications** and continuously learning emerging technologies to solve real-world challenges.

PROJECTS

Emotion-Aware Voice Processing for Customer Services (CNN-BiLSTM)

- Developed an **AI-powered** voice processing platform integrating **speech recognition**, **emotion detection**, noise reduction, and voice enhancement.
- Trained a custom **CNN-BiLSTM** model on the **RAVDESS** and **TESS** datasets for high-accuracy real-time speech emotion recognition.
- Designed scalable **FastAPI** services and optimized AI inference pipelines to deliver responsive voice analytics for customer service applications.

AI Content Assistant | Python, FastAPI, React, Groq Llama 3.3

- Developed a full-stack **AI content generation** platform integrating **Groq Llama 3.3**, FastAPI, and React for intelligent real-time text generation.
- Designed scalable **REST APIs** and optimized frontend-backend communication to deliver fast, reliable, and responsive **AI interactions**.
- Implemented clean architecture and **LLM integration** to generate accurate, context-aware responses while ensuring maintainable, production-ready code.

VizoMatch AI Movie Recommendation Engine | Python, Streamlit, Scikit-learn, TMDB API

- Developed a content-based **movie recommendation system** using **Machine Learning**, and **Streamlit** to deliver personalized movie suggestions.
- Built an NLP pipeline with **feature engineering**, vectorization, and cosine similarity to generate **accurate recommendations** from movie metadata.
- Integrated the **TMDB API** and optimized recommendation performance using precomputed **similarity matrices** for a fast, interactive user experience.

PROFESSIONAL EXPERIENCE

American Spaces Pakistan | Lincoln Corner Vehari Internhip (2024-2026)

- Organized and facilitated coding, **STEAM**, and **Generative AI workshops**, contributing to technical training and community outreach initiatives.
- Coordinated **educational programs**, managed **digital resources**, and promoted technology-driven learning through effective teamwork and communication.

Machine Learning Trainee | University of Education | 2024

- Applied machine learning techniques, including **Regression**, **Classification**, **Clustering**, **KNN**, **Decision Trees**, and **SVM**, to solve real-world analytical problems.
- Developed end-to-end machine learning workflows involving **data preprocessing**, **model training**, **evaluation**, and **performance optimization** using Python and Scikit-learn.

AI Engineering Intern | Decode Labs | 2026

- Developed AI and machine learning applications, including **intelligent chatbots**, **classification models**, and **recommendation systems** using Python and Scikit-learn.
- Applied **machine learning** and **deep learning techniques** to build, evaluate, and optimize real-world AI solutions through hands-on industrial projects.

TECHNICAL SKILLS

- Programming:** Python
- Machine Learning & AI:** Machine Learning, Deep Learning, NLP, LLMs, TensorFlow, Keras, Scikit-learn
- Data Science:** Data Analysis, Data Preprocessing, Feature Engineering, Pandas, NumPy
- Frameworks:** FastAPI, React, Streamlit
- Developer Tools:** Git, GitHub, Google Colab, Jupyter Notebook
- AI Tools:** Prompt Engineering, REST API Integration, Librosa

LANGUAGES

English | Urdu